

Assignment 3 - written

(1)

Lemma 3.1

Let $k \in \mathbb{N}_0$ and $s \in 2\mathbb{N} - 1$. Then it holds that for all $\varepsilon > 0$ there exists a shallow tanh neural network $\Psi_{s,\varepsilon} : [-M, M] \rightarrow \mathbb{R}^{\frac{s+1}{2}}$ of width $\frac{s+1}{2}$ such that

$$\max_{\substack{y \leq s, \\ p \text{ odd}}} \left\| f_p - (\Psi_{s,\varepsilon})^{\frac{p+1}{2}} \right\|_{W^{k,\infty}} \leq \varepsilon, \quad (17)$$

Moreover, the weights of $\Psi_{s,\varepsilon}$ scale as $O\left(\varepsilon^{-s/2} \left(2(s+2)\sqrt{2M}\right)^{s(s+3)}\right)$ for small ε and large s .

Core meaning

suppose have an odd-degree polynomial

$p(x) = x^s$ ($s = 1, 3, 5$), which is smooth (its derivative

are continuous). Lemma 3.1 states that there exist

a tanh-based network $\Psi_{s,\varepsilon}(x)$ with only one hidden layers

and enough neurons, such that error between $p(x)$ and

network output is less than any chosen $\varepsilon > 0$ and

if you want approach x^s , you will need

$s + \frac{1}{2}$ neurons.

Simple proof

start with Taylor Expansion

$$\tanh = x - \frac{1}{3}x^3 + \frac{2}{15}x^5 - \dots \cdot R_n(x)$$

by the note in class

Finite difference method (by 計算數學)

to approach $f(a)$, $f'(a)$

$$f(a+h) = f(a) + f'(a)h + \frac{f''(a)h^2}{2!} \dots$$

$$f(a-h) = f(a) - f'(a)h + \frac{f''(a)h^2}{2!} - \dots$$

$$\Rightarrow \underbrace{\frac{f(a+h) - f(a-h)}{2h} = f'(a) + \frac{f''(a)}{6}h^2 + O(h^4)}_{\text{center difference to approx } f'(x)}, \underbrace{\frac{f(a+h) - 2f(a) + f(a-h)}{h^2} = f''(a) + O(h^2)}_{\text{center difference approx } f''(a)}$$

↙ $\frac{3+1}{2}$

so two neuron $(\tanh(h), \tanh(-h))$ can

approximate $\tanh$ accurately to x^3

$\deg(R_n(x)) = 5$, and add one neuron (total 3 neurons)

we can approximate accurately to x^5 ,

↑
 $\frac{5+1}{2}$

so use same step, when we have

enough neuron, $R_n(x) \rightarrow 0$, so

$\forall \varepsilon$, $0 < R_n(x) < \varepsilon$, and to approximate

x^5 we need $\frac{5+1}{2}$ neuron.

Lemma 3.2 ,

Lemma 3.2

Let $k \in \mathbb{N}_0$, $s \in 2\mathbb{N} - 1$ and $M > 0$. For every $\varepsilon > 0$, there exists a shallow tanh neural network $\psi_{s,\varepsilon} : [-M, M] \rightarrow \mathbb{R}^s$ of width $\frac{3(s+1)}{2}$ such that

$$\max_{p \leq s} \|f_p - (\psi_{s,\varepsilon})_p\|_{W^{k,\infty}} \leq \varepsilon. \quad (26)$$

Furthermore, the weights scale as $O\left(\varepsilon^{-s/2} \left(\sqrt{M}(s+2)\right)^{3s(s+3)/2}\right)$ for small ε and large s .

like lemma 3.1, but this we want

approach any polynomial f_p ($p < s$)

so lemma 3.2 said that we can

approximate any f_p ($p < s$), too. but

we need more neurons to do it.

(Because we need to approximate the even order part, it may use some odd function to eliminate odd order part

to approximate the even part)

and if we want approach f_p ($p < s$)

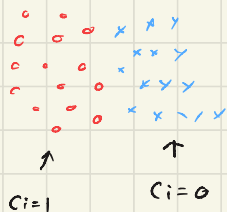
we need $\frac{3(n+1)}{2}$ neurons.

2,

by 上課筆記

Classification

data $\{(\vec{x}_i, c_i)\}$ $c_i \in [0, 1]$



Method 1: Find a $\text{func}: \mathbb{R}^2 \rightarrow \mathbb{R}$

s.t. $H(\vec{x}_i) = c_i$ (可能在交界學不好)

Method 2: One-hot encoding

$c_i \in \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$, Find a $\text{func} \mathbb{R}^2 \rightarrow \mathbb{R}^2$

s.t. $H(\vec{x}_i) = c_i$ (得勝說比較好 but don't know why)



老師是不是

還沒有說為什麼比較好